Syllabus Details

Syllabus ID:	13136
Syllabus Name:	**Statistics & Probability_Xác suất thống kê**
Course Name English:	
Subject Code:	**MAS291**
Learning-Teaching Method:	Lecture,Project based learning,Problem based learning,Social Constructivism,groupwork
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45 contact hours (60 sessions) + 1 hour final exam + 104 hours self-study
Pre-Requisite:	MAE101 or MAC101
Description:	Upon finishing the course, students must acquire: 1. the following knowledge:(ABET a1) • The fundamental principles of probability and their applications • The frequently used probability distributions. • The basics of descriptive statistics • The inferences of statistics: parameter estimations, hypothesis testing, regressions & correlations. 2. the following skills: (ABET a2) • Recognize simple statistical models and applied them to solve engineering problems. • Use at least one statistical software (Excel, Maxima) for problem solving. • Self-study skill. (ABET i)
StudentTasks:	Students must attend at least 80% of contact hours in order to be accepted to the final examination.
Tools:	- Internet - Computerized tools (for example Excel and DDXL add-ons) (CT)
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	932/QĐ-ĐHFPT dated 08/22/2025
IsApproved:	**True**
Note:	"On-going assessment: • 03 progress tests: 30% (a1, i) • 02 assignments 20% (a2) • 01 computer project 15% Final Examination 35% (a1, a2) Final results 100% Completion Criteria: Every on-going assessment component > 0, Final Result >=5 & Final Exam Score >=4 "
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/22/2025

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Montgomery D.C. & Runger G.C., Applied Statistics and Probability for Engineers	John Wiley & Sons.	Wiley	2006	4th/5th/6th/7th ed.	9781119718871	True	True	True	7th ed. ISBN: 9781119718871
2	Powerpoint slides (PP).						False	False	False	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	Computerized tools (for example Excel and DDXL add-ons) (CT)						False	False	False	
4	Elementary Statistics Using Excel, 3/E,	Mario F. Triola	Addison-Wesley	200	3rd	0321365135	False	True	False	

9 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Discuss the roles of statistics in engineering LO1.1 Explain the types of reasoning of statistical inference; distinguish mechanistic models and empirical models, statistic and parameter, sample and population. LO1.2 Describe the different methods that engineers use to collect data.
2	CLO2	Identify sample spaces and events, and use basic rules to calculate the probability of an event LO2.1 Describe the sample space and events of an experiment; distinguish continuous and discrete sample spaces; perform operations on events. LO2.2 Calculate the probability of an event in a discrete sample space; use addition and complementary rules to find probabilities. LO2.3 Calculate conditional probabilities; use multiplication rule, total probability rule and Bayes theorem to find probabilities. LO2.4 Determine if two events are independent; use independence to find probabilities. LO2.5 Recognize a random variable; distinguish continuous and discrete random variables.
3	CLO3	Describe the probability distribution and calculate the mean and variance of a discrete random variable LO3.1 Describe a discrete random variable; check if a function is a probability mass function and use it to calculate probability; find the cumulative distribution function of a discrete random variable. LO3.2 Compute the mean and variance of a discrete random variable. LO3.3 Determine the probability, mean and variance of uniform, binomial, geometric and negative binomial, hypergeometric and Poisson distributions.
4	CLO4	Describe the probability distribution and calculate the mean and variance of a continuous random variable LO4.1 Describe a continuous random variable; check if a function is a probability density function and use it to calculate probability; find the cumulative distribution function of a continuous random variable. LO4.2 Compute the mean and variance of a continuous random variables LO4.3 Apply the formulas of continuous uniform distribution and exponential distribution to solve problems. LO4.4 Manipulate probabilities of a standard normal distribution using Excel or table of data; standardize a normal distribution. LO4.5 Use normal distribution to approximate binomial and Poisson distributions.
5	CLO5	Describe data using numerical summaries and visual displays LO5.1 Find numeric summaries of data; describe data using stem-and-leaf diagrams, frequency distributions, histograms, time series plots. LO5.2 Determine quartiles and construct the box plot for a data; identify outliers.
6	CLO6	Discuss sampling distributions and use the Central Limit Theorem to analyze sampling distribution of the means LO6.1 Discuss sampling distributions; find the point estimation of a parameter. LO6.2 Apply Central Limit Theorem to calculate probabilities regarding the sampling distribution of the means.
7	CLO7	Construct confidence intervals for a population parameter/difference of population parameters LO7.1 Construct a confidence interval for the population mean/difference of two population means and interpret its meaning; determine the size of the sample to improve the estimation. LO7.2 Construct a confidence interval for the population proportion/difference of two population pproportions and interpret its meaning; determine the size of the sample to improve the estimation.
8	CLO8	Verify a claim about a population parameter/difference of population parameters using hypothesis testing LO8.1 Formulate the two hypotheses; identify critical values, test statistic and compute P-value for a test of hypotheses on population mean/difference of two population means LO8.2 Formulate the two hypotheses; identify critical values, test statistic and compute P-value for a test of hypotheses on population proportion/difference of two population proportions LO8.3 Identify type of errors in testing hypotheses.
9	CLO9	Construct a simple linear regression model LO9.1 Compute various sums of squares for a set of data pairs. LO9.2 Determine the equation of linear regression and the correlation coefficient; use linear regression to predict future values. LO9.3 Perform a hypothesis test in simple linear regression; test the significance of regression using t-test.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	The Roles of Statistics in Engineering 1.1 The Engineering Method and Statistical Thinking 1.2 Collecting Engineering Data	Offline	LO1	IT	Textbook Chapter 1		Exercises Chapter 1 Assignment 1	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
2	The Roles of Statistics in Engineering 1.3 Mechanistic and Empirical Models 1.4 Probability and Probability Models	Offline	LO1	IT	Textbook Chapter 1	MAS291	Exercises Chapter 1 Assignment 1	
3	Probability 2.1 Sample Spaces and Events	Offline	LO2	IT	Textbook Chapter 2		Exercises Chapter 2 Assignment 1	
4	Probability 2.2 Interpretations of Probability 2.3 Addition Rules	Offline	LO2	IT	Textbook Chapter 2		Exercises Chapter 2 Assignment 1	
5	Probability 2.4 Conditional Probability	Offline	LO2	TU	Textbook Chapter 2		Exercises Chapter 2 Assignment 1	
6	Probability 2.5 Multiplication and Total Probability Rules 2.6 Independence	Offline	LO2	TU	Textbook Chapter 2		Exercises Chapter 2 Assignment 1	
7	Probability 2.7 Bayes' Theorem	Offline	LO2	ITU	Textbook Chapter 2		Exercises Chapter 2 Assignment 1	
8	Probability 2.8 Random Variables	Offline	LO2	ITU	Textbook Chapter 2		Exercises Chapter 2 Assignment 1	
9	Discrete Random Variables and Probability Distribution 3.1 Discrete Random Variables	Offline	LO3	ITU	Textbook Chapter 3		Exercises Chapter 3 Assignment 1	
10	Discrete Random Variables and Probability Distribution 3.2 Probability Distributions and Probability Mass Functions	Offline	LO3	ITU	Textbook Chapter 3		Exercises Chapter 3 Assignment 1	
11	Discrete Random Variables and Probability Distribution 3.3 Cumulative Distribution Functions	Offline	LO3	TU	Textbook Chapter 3		Exercises Chapter 3 Assignment 1	
12	Discrete Random Variables and Probability Distribution 3.4 Mean and Variance of a Discrete Random Varibale	Offline	LO3	TU	Textbook Chapter 3		Exercises Chapter 3 Assignment 1	
13	Discrete Random Variables and Probability Distribution 3.5 Discrete Uniform Distribution 3.6 Binomial Distribution	Offline	LO3	TU	Textbook Chapter 3		Exercises Chapter 3 Assignment 1	
14	Discrete Random Variables and Probability Distribution 3.7 Geometric and Negative Binomial Distributions	Offline	LO3	TU	Textbook Chapter 3		Exercises Chapter 3 Assignment 1	
15	Discrete Random Variables and Probability Distribution 3.8 Hypergeometric Distribution	Offline	LO3	TU	Textbook Chapter 3		Exercises Chapter 3 Assignment 1	
16	Discrete Random Variables and Probability Distribution 3.9 Poisson Distribution	Offline	LO3	TU	Textbook Chapter 3		Exercises Chapter 3 Assignment 1	
17	Solve questions and problems for Chapters 1 - 3	Offline	LO1, LO2, LO3		Textbook Chapters 1 - 3		Exercises Chapters 1 - 3 Assignment 1	
18	Progress Test 1	Offline	LO1, LO2, LO3					

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
19	Continuous Random Variables and Probability Distribution 4.1 Continuous Random Variables 4.2 Probability Distributions and Probability Density Functions	Offline	LO4	ITU	Textbook Chapter 4		Exercises Chapter 4 Assignment 2	
20	Continuous Random Variables and Probability Distribution 4.3 Cumulative Distribution Functions	Offline	LO4	ITU	Textbook Chapter 4		Exercises Chapter 4 Assignment 2	
21	Continuous Random Variables and Probability Distribution 4.4 Mean and Variance of a Continuous Random Variable	Offline	LO4	TU	Textbook Chapter 4		Exercises Chapter 4 Assignment 2	
22	Continuous Random Variables and Probability Distribution 4.5 Continuous Uniform Distribution	Offline	LO4	TU	Textbook Chapter 4		Exercises Chapter 4 Assignment 2	
23	Continuous Random Variables and Probability Distribution 4.6 Normal Distribution	Offline	LO4	TU	Textbook Chapter 4		Exercises Chapter 4 Assignment 2	
24	Continuous Random Variables and Probability Distribution 4.6 Normal Distribution (Cont.)	Offline	LO4	TU	Textbook Chapter 4		Exercises Chapter 4 Assignment 2	
25	Continuous Random Variables and Probability Distribution 4.7 Normal Approximation to the Binomial and Poisson Distributions	Offline	LO4	TU	Textbook Chapter 4		Exercises Chapter 4 Assignment 2	
26	Continuous Random Variables and Probability Distribution 4.8 Exponential Distribution	Offline	LO4	TU	Textbook Chapter 4		Exercises Chapter 4 Assignment 2	
27	Descriptive Statistics 6.1 Numerical Summaries	Offline	LO5	IT	Textbook Chapter 6		Exercises Chapter 6 Assignment 2	
28	Descriptive Statistics 6.2 Stem-and-Leaf Diagrams 6.3 Frequency Distributions and Histograms	Offline	LO5	IT	Textbook Chapter 6		Exercises Chapter 6 Assignment 2	
29	Descriptive Statistics 6.4 Box Plots	Offline	LO5	IT	Textbook Chapter 6		Exercises Chapter 6 Assignment 2	
30	Descriptive Statistics 6.5 Time Sequence Plots	Offline	LO5	IT	Textbook Chapter 6		Exercises Chapter 6 Assignment 2	
31	Sampling Distributions and Point Estimation of Paramaters 7.1 Point Estimation	Offline	LO6	I	Textbook Chapter 7		Exercises Chapter 7 Assignment 2	
32	Sampling Distributions and Point Estimation of Paramaters 7.2 Sampling Distributions and the Central Limit Theorem	Offline	LO6	I	Textbook Chapter 7		Exercises Chapter 7 Assignment 2	
33	Solve questions and problems for Chapters 4, 6, and 7	Offline	LO4, LO5, LO6		Textbook Chapters 4, 6, and 7		Exercises Chapters 4, 6, and 7 Assignment 2	
34	Progress Test 2	Offline	LO4, LO5, LO6					

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
35	Statistical Intervals for a Single Sample 8.1 Confidence Interval on the Mean of a Normal Distribution, Variance Known	Offline	LO7	TU	Textbook Chapter 8		Exercises Chapter 8 Project	
36	Statistical Intervals for a Single Sample 8.1 Confidence Interval on the Mean of a Normal Distribution, Variance Known (Cont.)	Offline	LO7	TU	Textbook Chapter 8		Exercises Chapter 8 Project	
37	Statistical Intervals for a Single Sample 8.2 Confidence Interval on the Mean of a Normal Distribution, Variance Unknown	Offline	LO7	TU	Textbook Chapter 8		Exercises Chapter 8 Project	
38	Statistical Intervals for a Single Sample 8.2 Confidence Interval on the Mean of a Normal Distribution, Variance Unknown (Cont.)	Offline	LO7	TU	Textbook Chapter 8		Exercises Chapter 8 Project	
39	Statistical Intervals for a Single Sample 8.4 Large-Sample Confidence Interval for a Population Proportion	Offline	LO7	TU	Textbook Chapter 8		Exercises Chapter 8 Project	
40	Statistical Intervals for a Single Sample 8.4 Large-Sample Confidence Interval for a Population Proportion (Cont.)	Offline	LO7	TU	Textbook Chapter 8		Exercises Chapter 8 Project	
41	Test of Hypotheses for a Single Sample 9.1 Hypothesis Testing	Offline	LO8	TU	Textbook Chapter 9		Exercises Chapter 9 Project	
42	Test of Hypotheses for a Single Sample 9.1 Hypothesis Testing (Cont.)	Offline	LO8	TU	Textbook Chapter 9		Exercises Chapter 9 Project	
43	Test of Hypotheses for a Single Sample 9.2 Tests on the Mean of a Normal Distribution, Variance Known	Offline	LO8	IT	Textbook Chapter 9		Exercises Chapter 9 Project	
44	Test of Hypotheses for a Single Sample 9.2 Tests on the Mean of a Normal Distribution, Variance Known (Cont.)	Offline	LO8	IT	Textbook Chapter 9		Exercises Chapter 9 Project	
45	Test of Hypotheses for a Single Sample 9.3 Tests on the Mean of a Normal Distribution, Variance Unknown	Offline	LO8	TU	Textbook Chapter 9		Exercises Chapter 9 Project	
46	Test of Hypotheses for a Single Sample 9.5 Tests on a Population Proportion	Offline	LO8	TU	Textbook Chapter 9		Exercises Chapter 9 Project	
47	Statistical Inference for Two Samples 10.1 Inference on the Difference in Means of Two Normal Distributions, Variance Known	Offline	LO7,LO8	TU	Textbook Chapter 10		Exercises Chapter 10 Project	
48	Statistical Inference for Two Samples 10.2 Inference on the Difference in Means of Two Normal Distributions, Variance Unknown	Offline	LO7,LO8	TU	Textbook Chapter 10		Exercises Chapter 10 Project	
49	Statistical Inference for Two Samples 10.6 Inference on the Two Proportions	Offline	LO7,LO8	TU	Textbook Chapter 10		Exercises Chapter 10 Project	
50	Statistical Inference for Two Samples 10.6 Inference on the Two Proportions (Cont.)	Offline	LO7,LO8	TU	Textbook Chapter 10		Exercises Chapter 10 Project	
51	Simple Linear Regression and Correlation 11.1 Empirical Models	Offline	LO9	I	Textbook Chapter 11		Exercises Chapter 11 Project	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
52	Simple Linear Regression and Correlation 11.2 Simple Linear Regression	Offline	LO9	I	Textbook Chapter 11		Exercises Chapter 11 Project	
53	Simple Linear Regression and Correlation 11.3 Properties of the Least Squares Estimators	Offline	LO9	ITU	Textbook Chapter 11		Exercises Chapter 11 Project	
54	Simple Linear Regression and Correlation 11.4 Hypothesis Tests in Simple Linear Regression	Offline	LO9	ITU	Textbook Chapter 11		Exercises Chapter 11 Project	
55	Simple Linear Regression and Correlation 11.8 Correlation	Offline	LO9	TU	Textbook Chapter 11		Exercises Chapter 11 Project	
56	Simple Linear Regression and Correlation 11.8 Correlation (Cont.)	Offline	LO9	TU	Textbook Chapter 11		Exercises Chapter 11 Project	
57	Project presentation	Offline	LO7, LO8, LO9	U			Project	
58	Project presentation (Cont.)	Offline	LO7, LO8, LO9	U			Project	
59	Project presentation (Cont.)	Offline	LO7, LO8, LO9	U			Project	
60	Progress Test 3	Offline	LO7, LO8, LO9	U				

111 Constructive question(s)

No.	Session No	Name	Details
1	1	CQ1_DN	Statistics is the science of data. What are the basic data collection methods?
2	1	CQ1_HCM	What are parameters of the population and statistics of a sample?
3	1	CQ2_HCM	List some commonly used parameters and statistics.
4	2	CQ1_DN	An event is a subset of the sample space of a random experiment. Can we describe new events from combinations of existing events?
5	2	CQ1_HCM	What are the differences between empirical and mechanistic models?
6	3	CQ1_DN	Counting the numbers of outcomes in the sample space and various events are used to analyze the random experiments. What are important techniques?
7	3	CQ1_HCM	Give two different definitions of the concept of events.
8	4	CQ1_DN	For discrete sample spaces S consisting of (equally likely) outcomes, the probability of an event E (a subset of S) equals the sum of the probabilities of the outcomes in E. Can we determine the probability of a joint event from the probabilities of the individual events that comprise it?
9	4	CQ1_HCM	What are independent events and mutually exclusive events? How do they affect the formula for computing the probabilities of the union event, the intersection event?
10	5	CQ1_DN	Sometimes probabilities need to be reevaluated as additional information becomes available. Can you give the definition of conditional probabilities and some examples?
11	5	CQ1_HCM	How to check whether two events are independent?
12	6	CQ1_DN	Sometimes the probability of an event is given under each of several conditions. With enough of these conditional probabilities, how to recover the probability of the event?
13	6	CQ1_HCM	What conditions must be fulfilled to be able to use the total probability rule?
14	6	CQ2_HCM	Give an example that uses the rule of total probability.

No.	Session No	Name	Details
15	7	CQ1_DN	Conditional probabilities commonly provide the probability of an event given a condition. But after a random experiment generates an outcome, can we find out the probability that a condition was present given an outcome?
16	7	CQ1_HCM	Why is Bayes Theorem important for probability?
17	8	CQ1_DN	The concept of independence is an important relationship between events. Can you give the definition of independence and some examples?
18	8	CQ1_HCM	What is a random variable? Give some examples.
19	9	CQ1_DN	The probability distribution of a discrete random variable X is a description of the probabilities associated with the possible values of X. How many popular ways to describe a probability distribution?
20	9	CQ1_HCM	Compare the difference between discrete and continuous variables.
21	9	CQ2_HCM	Give some examples of discrete and continuous variables.
22	10	CQ1_HCM	What is a probability mass function?
23	10	CQ2_HCM	What is the difference between a probability mass function and a cumulative mass function?
24	11	CQ1_HCM	How do you find the probability mass function if the cumulative distribution function is given, and vice versa? Give an example to illustrate your claim.
25	12	CQ1_DN	What is the simplest discrete random variable? Can we numerically summarize its probability distribution?
26	12	CQ1_HCM	Give a table of data showing a car rental company's revenue over the year, what do the expected value and variance represent?
27	13	CQ1_DN	How to calculate the probability of k successes in a random experiment consists of n Bernoulli trials?
28	13	CQ1_HCM	Give an example of a binomial random variable that you have seen in high school.
29	14	CQ1_DN	Is negative binomial distribution a generalization of a geometric distribution? Why do we call this distribution a negative binomial distribution?
30	14	CQ1_HCM	How to distinguish the differences among a binomial, geometric and negative binomial random variable?
31	14	CQ2_HCM	Give examples of biomial, geometric, and hypergeometric distributions.
32	15	CQ1_DN	In randomly sampling from a batch of items classified as successes or failures, an item may or may not be replaced before the next one is selected. Does it exist or not a general formula for computing probabilities when samples are selected without replacement?
33	15	CQ1_HCM	When is a random variable called a hypergeometric variable? Show an example.
34	16	CQ1_DN	A widely-used distribution emerges as the number of trials in a binomial experiment (with parameters n, p) increases to infinity while the mean of the distribution remains constant (n increases and p decreases accordingly). What is this?
35	16	CQ1_HCM	What properties are required to use the Poisson distribution?
36	16	CQ2_HCM	Give two real life examples for which Poisson distributions are used as models.
37	17	CQ1_HCM	What are the properties required to use the binomial distribution?
38	18	CQ1_HCM	What is the meaning of the expected value of a variable?
39	18	CQ2_HCM	Explain how to compute expected value of a random variable by examples.
40	19	CQ1_DN	Because the number of possible values of a continuous random variable X is uncountably infinite, X has a distinctly different distribution from the discrete random variables studied previously. How to describe the probability distribution of X?
41	19	CQ1_HCM	What makes a function a density function?
42	20	CQ1_HCM	What is the cumulative distribution function of a continuous random variable and its properties?
43	21	CQ1_DN	The simplest continuous distribution is analogous to its discrete counterpart. Give some information about it?
44	21	CQ1_HCM	How do you find the mean of a continuous random variable?
45	21	CQ2_HCM	How do you find the variance of a continuous random variable?
46	22	CQ1_DN	Probabilities of intervals can be found by using the cumulative distribution function, the basic rules of probability and the symmetry of the normal distribution. Can you illustrate the method by some examples?

No.	Session No	Name	Details
47	22	CQ1_HCM	What are the characteristics of a continuous uniform distribution?
48	23	CQ1_DN	We know how to calculate probabilities for standard normal random variables. Can we apply the method to find the probabilities associated with an arbitrary normal random variable?
49	23	CQ1_HCM	What are the main properties of normal distribution?
50	23	CQ2_HCM	What is the rule called '68-95-99 rule'?
51	24	CQ1_DN	For cases in which the number of trials is large, it is difficult to calculate probabilities by using the binomial distribution. How do statisticians deal with this problem?
52	24	CQ1_HCM	How could you find the area between two values under the normal curve?
53	25	CQ1_DN	It is known that, there are intimate connections between three distributions of binomial, hypergeometric and Poisson. What are they? Is it possible to use them to approximate hypergeometric and binomial probabilities?
54	25	CQ1_HCM	What is the difference between a normal distribution and a binomial distribution?
55	26	CQ1_DN	The Poisson distribution defines a random variable to be the number of events occurring in a fixed interval of time if these events occur with a known constant mean rate and independently of time since the last event. Can we describe the probability distribution of the time between those events?
56	26	CQ1_HCM	What is the exponential distribution?
57	26	CQ2_HCM	Where is the exponential distribution used in the real world?
58	27	CQ1_DN	An important aspect of dealing with data is organizing and summarizing the data. How should we numerically summarize the information in these data?
59	27	CQ1_HCM	List some ways to measure central tendency. What are the formulas?
60	28	CQ1_DN	The graphical techniques are very useful in constructing and interpreting visual data displays. What are most popular techniques?
61	28	CQ1_HCM	How do you construct a table of frequency distribution and the corresponding histogram? Give an illustrative example.
62	29	CQ1_HCM	Is a Box plot used for outlier detection? How do you find them, if possible?
63	30	CQ1_HCM	Give an example of time series.
64	30	CQ2_HCM	What are the applications of time series analysis in practice?
65	31	CQ1_DN	One of the most useful theorems in statistics called the central limit theorem. What is this?
66	31	CQ1_HCM	What is an unbiased and biased estimator? Give an example.
67	32	CQ1_DN	How to use the central limit theorem to determine the point estimate for the proportion?
68	32	CQ1_HCM	What is the difference between a population distribution and a sampling distribution?
69	33	CQ1_HCM	Why does the standard error of the sample mean decrease as the sample size, n, increases?
70	33	CQ2_HCM	What is the formula for calculating the standard error of the sample mean when the sample size is n?
71	34	CQ1_HCM	In what conditions is the sampling distribution of the sample mean approximately normal?
72	35	CQ1_DN	There is nothing inherent in the sample mean that provides any information about how close it is to the unknown true mean. Can we produce the interval that contains the true mean of a normal distribution, variance known?
73	35	CQ1_HCM	Why don't we use 100% confidence in correctly estimating the population parameters?
74	36	CQ1_HCM	What is the difference between a confidence interval and a confidence level?
75	36	CQ2_HCM	Analyze the influence of increasing the confidence level to the width of the corresponding confidence interval.
76	37	CQ1_DN	Check the affirmation that "The general appearance of the t distribution is similar to the standard normal distribution."
77	37	CQ1_HCM	What is the t-distribution?
78	37	CQ2_HCM	When should you use the t-distribution to design a confidence interval estimate for the mean?
79	38	CQ1_DN	How to construct confidence intervals on the mean of a normal distribution when the sample is small and its variance is unknown?

No.	Session No	Name	Details
80	38	CQ1_HCM	How would you interpret a 95% confidence interval for the mean?
81	39	CQ1_HCM	When determining the required sample size for a given confidence level and the error, how does the sample size change as we increase/decrease the population proportion?
82	40	CQ1_HCM	For a fixed sample size, increasing the confidence level will widen (or make it less precise) the confidence interval. Show your opinion about this statement.
83	41	CQ1_HCM	What does P-value mean?
84	42	CQ1_DN	What is the difference between two approaches used in hypothesis testing: critical value and p-value?
85	42	CQ1_HCM	What is the difference between a Type I error and a Type II error?
86	42	CQ2_HCM	How do we calculate a type II error?
87	43	CQ1_HCM	What is the difference between a one-tailed test and a two-tailed test?
88	44	CQ1_HCM	What is the test statistic on the mean of a normal distribution with known variance?
89	44	CQ2_HCM	What is formula for finding the test statistic on the mean of a normal distribution with known variance?
90	45	CQ1_HCM	When do you use the t-test for testing a hypothesis?
91	46	CQ1_HCM	How can you use the p-value approach to the Z-test of the hypothesis for the proportion?
92	47	CQ1_HCM	Which test can be used for the difference between the two means? Why?
93	47	CQ2_HCM	Give an example of testing the difference between the two means.
94	48	CQ1_HCM	Under what conditions should you use the pooled-variance to test the differences in the means of two independent populations?
95	49	CQ1_HCM	What are the assumptions for the two samples' proportion test?
96	50	CQ1_HCM	In two independent populations, what are the similarities and differences between the hypothesis test and the confidence interval estimate for the difference between two proportions?
97	51	CQ1_HCM	What is the problem setting of simple linear regression?
98	51	CQ2_HCM	Find some applications of regression in the real world.
99	52	CQ1_HCM	What is a dependent and independent variable? Explain them with examples.
100	53	CQ1_HCM	What is the interpretation of the Y intercept and the slope in the simple linear regression equation?
101	54	CQ1_HCM	What are the required assumptions of regression analysis?
102	54	CQ2_HCM	Explain the meaning of the word 'linear' in the concept 'linear regression'.
103	55	CQ1_DN	How to determine whether there is a significant linear relationship between two variables in the population?
104	55	CQ1_HCM	What is the difference between the sample correlation coefficient and coefficient of determination?
105	55	CQ2_HCM	Explain the meaning of R^2?
106	56	CQ1_HCM	Which test statistic is used to test the relationship between two variables?
107	57	CQ1_HCM	Using the Wage dataset (education on wages) from 57 to 60. Classify the following variable: nearc4, educ, age, black, wage, IQ, married, exper, lwage, and expersq.
108	58	CQ1_HCM	What are the coefficients slope and intercept?
109	58	CQ2_HCM	Run regression equation of educ on wage, interpret the slope and the intercept from the output.
110	59	CQ1_HCM	Suppose that you examine two models: educ on wage and educ on lwage. Which model is better? Why?
111	60	CQ1_HCM	You are interested in studying factors affecting wages. Which models should you recommend?

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	2	20.0%	>0	3 hours	LO1, LO2, LO3, LO4, LO5, LO6	OP1: Open // OP2: (For Constructivism Approach only): Follow lecturer's proposal	5--10	Assignment 1 covers Chapters 1, 2, 3 ; Assignment 2 covers Chapters 4, 6, 7	OP1: Each assignment is divided into two parts: the take-home component counts for no more than 50% of the total grade, and the in-class component counts for at least 50% (this may take the form of a presentation, essay, problem-solving, etc.). OP2: (For Constructivism Approach only): Follow lecturer's proposal	Assignment 1: LO1, LO2, LO3 Assignment 2: LO4, LO5, LO6

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
2	Computer Project	on-going	1	15.0%	>0	5 hours	LO7, LO8, LO9	OP1: Open // OP2: (For Constructivism Approach only): Follow lecturer's proposal	N/A	Computer Project covers Chapter 8, 9, 10, 11	OP1: Groups of students are required to collect secondary data (topics may be suggested by instructor), form their own questions, and apply learned knowledge and methods of inferential statistics to analyze data and make conclusions about the populations. Groups are expected to use a statistical software to perform the analysis and present their results to other groups in class. The work (slides, source files, etc.) accounts for up to 50% of the grade, while presentation skills and the ability to answer questions account for at least 50%. OP2: (For Constructivism Approach only): Follow lecturer's proposal	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
3	Progress Test	on-going	3	30.0%	>0	30'	All LOs	OP1: MC or a suitable format // OP2: (For Constructivism Approach only): Follow lecturer's proposal	15	Chapters introduced	OP1: by instructor, and by suitable means (computer, paper, oral...) OP2: (For Constructivism Approach only): Follow lecturer's proposal	OP1: Instruction and schedules for Progress tests must be presented in the Course Implementation Plan approved by director of the campus. Test 1: LO1, LO2, LO3 Test 2: LO4, LO5, LO6 Test 3: LO7, LO8, LO9 OP2: (For Constructivism Approach only): Follow lecturer's proposal Progress test must be taken right after the last lectures of required material. Instructor has resposibility to review the test for students after graded.
4	Final exam	Final exam	1	35.0%	4	60'		Computer gradable	50	All studied chapters 3-10 questions/each. More than 70% new questions (for the current semester);	by Exam Board	10% of questions is ABET a2-objective oriented